Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents with Turtle Soup Puzzles

Xiaoxuan Lei (co-authors TBD)

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Abstract

When an LLM agent fails a lateral-thinking task, outcome metrics report *that* it failed—not *how*. We present a behavioral-interpretation framework built on Turtle Soup (海龟汤, situation puzzles), an interactive game whose yes/no questioning protocol externalizes an agent's hypothesis-space search, turn by turn. Our central hypothesis is that lateral-thinking failures are legible in the geometry of the interaction trace: embedding each round's question into a semantic space anchored by human word-association norms separates two failure modes that identical outcome scores conflate—*stalling* (association step size collapsing toward zero) and *drifting* (normal step size but increasing distance from human association paths). We instantiate the framework in an open-source dual-agent harness, contribute a composite score that decouples objective clue recall from judged causal-logic recovery, and run three experiments over 11 puzzles $\times$ 3 model sizes $\times$ 3 seeds (693 games): round-curve (E1) and round-budget (E2) studies, and an association-trajectory study (E3) adapting forward-flow and divergent-association psychometrics to agent traces. Outcome curves are strikingly flat—checkpoint accuracy does not improve over 30 rounds for any model, and added budget leaves large models unchanged while actively degrading the 4B—and the geometry explains the flatness: drift away from the human manifold predicts failure ($r = -0.28$ pooled, -0.43 mid-size), while stride's sign *inverts with scale* (the small model's large jumps are noise, $\rho = -0.46$; the largest model's are productive exploration). We also report a methodological trap for interactive evaluation at large: a weak Oracle silently converts the task into one that is information-theoretically unsolvable, so observed "agent failure" can be environment failure.

1 Introduction

Benchmarks for agentic reasoning overwhelmingly score outcomes: solve rate, accuracy, rounds. For *lateral* thinking—abductive reconstruction of a hidden explanation from sparse, surprising observations—outcomes are especially uninformative: two agents scoring 0.0 may be failing in opposite ways (one never leaves its first hypothesis; one leaves the plausible region entirely). Interpreting agent behavior requires instruments that read the *process*.

Turtle Soup gives the agent a paradoxical scenario (the 汤面, "surface") and hides the story (the 汤底, "base"); the agent asks yes/no questions to reconstruct it. Three properties make it a behavioral microscope: (i) the questioning protocol is a naturally verbalized search trace—no activation access needed, the behavior *is* text; (ii) puzzles are deliberately under-determined, so the direction of each associative jump is diagnostic; (iii) games are short, language-only, and cheap at scale. Prior turtle-soup benchmarks score verification correctness, solve efficiency, or consistency with the intended story (§2); none interprets the search behavior itself.

Central hypothesis. An agent's per-round questions, embedded jointly with human word-association norms, form a trajectory whose geometry predicts and explains success:

- **H1 (stalling).** Failure trajectories with step size $\to 0$ identify agents that cannot leave a hypothesis basin; lexical novelty heuristics detect only the verbatim extreme.
- **H2 (drifting).** Failure trajectories with normal step size but increasing distance from the human association manifold identify agents exploring where no human path goes—invisible to lexical novelty, since synonym-shuffled re-asks look "new."
- **H3 (predictivity).** Trajectory features predict final solve quality beyond outcome-adjacent covariates, and the two failure modes have distinct signatures.

Contributions. (1) A behavioral-interpretation framework and open-source dual-agent harness; (2) a composite outcome score decoupling clue recall from judged causal logic, exposing when surface metrics mislead; (3) experiments E1–E3 with results on a 3-size model sweep; (4) a methodological finding: Oracle competence must be audited first, else agent scores measure the environment.

2 Related Work

Turtle soup as LLM benchmark. TurtleBench [1] collects 1,532 real user guesses and evaluates models as yes/no judges—verification, not exploration. SPLAT [2] introduces 975 graded situation puzzles and the interactive player–judge protocol we adopt, scored by accuracy and rounds. TurtleSoup-Bench [3] frames 800 bilingual puzzles as "imaginative reasoning" with multi-dimensional scoring, still rewarding faithfulness to the single intended story. Our delta: we score the *trajectory*, not only the destination.

Divergent thinking in agents. MUTATE [4] shows frontier agents discover far fewer mechanism-distinct alternative paths than humans, and localizes a cause: divergent generation and convergent selection under one conditioning. Its diverge-then-narrow agent (ReDNA) is a candidate intervention to test against our metrics: does it lengthen steps (H1) or re-anchor toward human paths (H2)? Cross-domain creativity evaluations [5, 6] supply the novelty$\times$appropriateness consensus our composite score implements.

Human creativity psychometrics we adapt. Forward flow [9] quantifies how far free-association chains travel in semantic space and predicts human creativity; the Divergent Association Task [10] shows semantic distance among generated words is itself a robust creativity measure, applied to LLMs by [12]. The Small World of Words norms [11] give an empirical human association manifold to anchor "human-path distance," rather than assuming any model embedding is human-like.

LLM-as-judge reliability. Bias catalogues [7, 8] motivate our judge design: rubric-anchored pointwise scoring, blinding, and confining the judge to the one subscore (causal-logic identity) that objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 and asks one yes/no question per round; an **Oracle** holds the 汤底 and answers 是/不是/与此无关; the Questioner may commit a final story at any round, or is forced to at the budget. The open-source harness has pluggable providers (local, gateway, and token-billed sampling of large open models, including RL-trained checkpoints), full trajectory logging, and per-game seeds.

3.2 Outcome scoring: composite, not surface

Judged outcome $=$ **clue recall (70)** $+$ **causal-logic identity (30)**. Clue recall matches per-puzzle annotated key clues (with a character-bigram-recall fallback for sentence-form clues; §6); the logic judge rates only whether cause$\rightarrow$mechanism$\rightarrow$outcome matches, ignoring wording, sampled k times and averaged. Distance without validity scores zero, per the novelty$\times$appropriateness consensus.

3.3 Association-trajectory instrumentation

Per round t: extract question keywords (fixed extractor), embed, aggregate to a unit vector q_t. Signals: **step size** $s_t = d(q_t, q_{t-1})$ (forward-flow analogue) and **human-path distance** $h_t = d(q_t, \mathcal{H})$, where $\mathcal{H}$ is the puzzle's human association anchor—SWOW-seeded in the full design; a documented proxy (surface/solution/clue content words) in this draft. Hypothesized signatures: stalling $= \bar{s} \rightarrow 0$, h flat; drifting $= \bar{s}$ normal, h_t increasing.

4 Experiments

Setup. 11 Chinese puzzles with annotated key clues; Questioners Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B (sampled via a token-billed API for open models); Oracle and logic judge fixed to Qwen3.5-397B-A17B, thinking disabled. **Oracle audit (prerequisite):** on held-out probe pairs the Oracle scores 15/15 (100%) on yes/no items. In an earlier pilot, a 4B Oracle answered 30% correctly, collapsing most questions to "irrelevant"—after which no Questioner can succeed; raising Oracle accuracy 30% $\rightarrow$ 100% left the 4B Questioner's score unchanged: the environment, not the agent, was the bottleneck.

E1 (round curve). Play to 30 rounds; after each round force a checkpoint answer, scored by clue recall. 11 puzzles $\times$ 3 models $\times$ 3 seeds $=$ 99 games. **E2 (round budget).** Hard caps $\{5, 10, 15, 20, 25, 30\}$ with forced final answer, composite scoring with logic rater; 594 games. **E3 (association trajectory).** For every E1 game: $\{s_t\}$, $\{h_t\}$; correlate trace geometry with outcome.

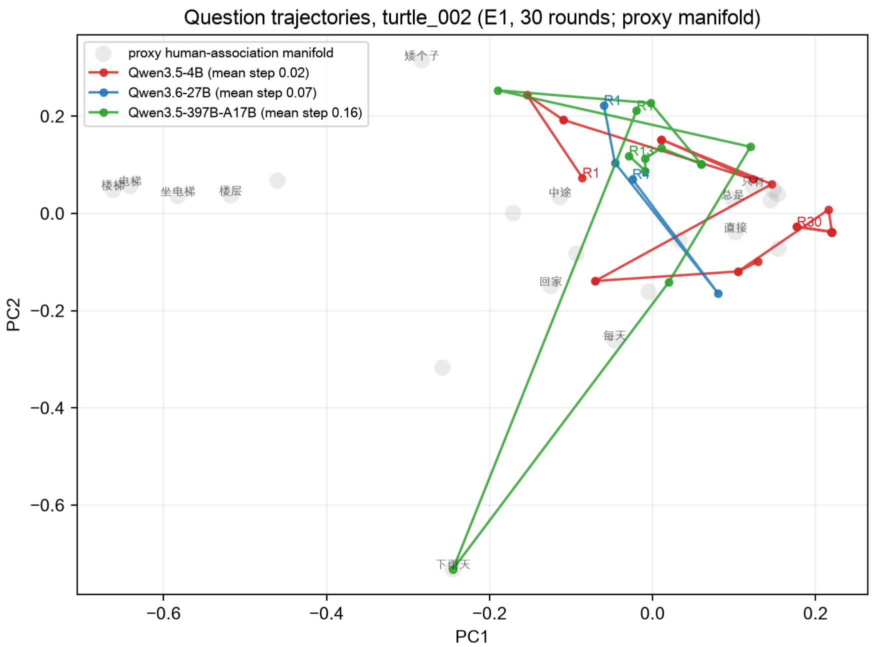

Figure 1: Question trajectories of three model sizes on one puzzle (E1, seed 0; 2-D PCA; gray cloud: proxy association manifold). The 4B (red) plays all 30 rounds with late-game step size 0.003—near-total stall. The 397B (green) strides at 0.16 for 13 rounds, narrowing to 0.02 as it converges—explore-then-commit. The 27B (blue) lands on the correct hypothesis at round 1 and commits by round 4: stride requires convergence context (§5).

5 Results

E1: interaction does not improve the working hypothesis. Checkpoint accuracy is flat across all 30 rounds for every model (Fig. 2, left): 27B ≈ 0.20, 397B ≈ 0.18, 4B ≈ 0.14; SEM bands overlap between round 1 and round 30 throughout. Commitment behavior, however, separates sharply: the 4B *never* voluntarily commits a final answer (0/33 games), the 27B commits readily (9/33, mean round 8.7), the 397B rarely and late (2/33, mean round 19.5)—the calibration axis of the behavioral profile discriminates even where accuracy does not.

E2: budget buys the small model degradation, not accuracy. The 27B and 397B are flat across caps (0.32–0.36; all 198 games each end in a committed answer under forcing). The 4B peaks at cap 15 (0.23) then declines monotonically to 0.10 at cap 30, with 29/198 games dying by token-budget exhaustion (Fig. 2, right). No model converts additional rounds into accuracy; the smallest converts them into noise.

E3: geometry explains the flatness, capacity-dependently. Over the 99 E1 traces: *drift predicts failure* (H2, direction confirmed vs. the proxy manifold)—human-distance slope vs. best accuracy $r = -0.28$ pooled, -0.43 for the 27B. *Stride is capacity-dependent* (H1 refined): the pooled stride–outcome correlation is ≈ 0, but per model it inverts with scale—4B $\rho = -0.46$ (large jumps are incoherent), 397B $r = +0.21$ (large jumps are productive exploration). Raw step size is not a univariate health signal; step size conditioned on capability tier is. At the trace level the signatures are vivid (Fig. 1). H3 (predictivity beyond covariates) is partially supported—drift adds signal, stride only in interaction with model tier; a mixed-effects analysis remains before claiming it fully.

Interpretation. The flat outcome curves are the *outcome shadow* of two different behaviors the geometry separates: small models stall (and degrade with budget); larger models explore but do not *bank* their exploration into an improving hypothesis. That diagnosis is invisible to every outcome metric and is the case for trajectory-level interpretation.

6 Pilot Observations

(1) **Stalling exists.** A 4B model re-asks earlier questions verbatim in later rounds—the lexical extreme of step-size collapse. (2) **Surface scoring misleads.** A frontier model reconstructed a hidden story's full causal chain yet scored 0.00 under clue-string matching; the composite score separated its logic (30/30) from its terminology (28/70). Relatedly, sentence-form clues degenerate token matching to exact substring—a correct paraphrase scored 0.00 until a

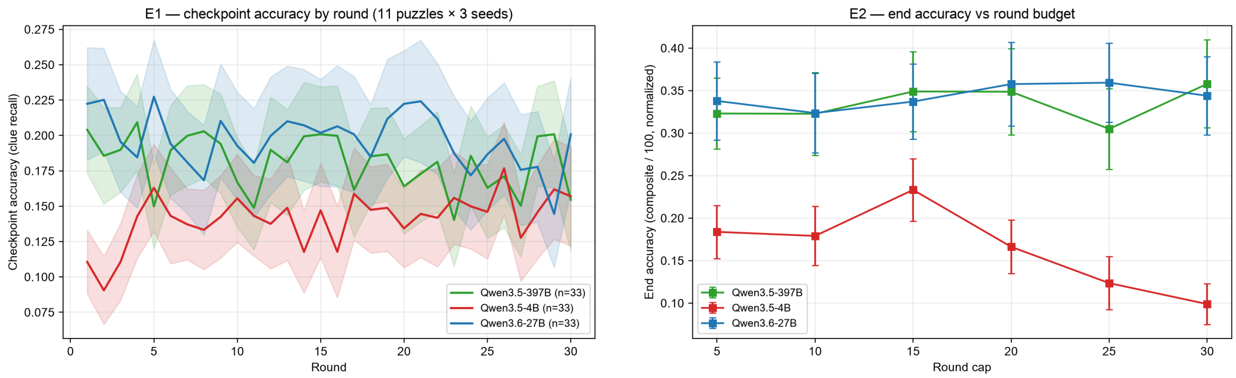

Figure 2: Left (E1): checkpoint accuracy is flat across 30 rounds for all models (mean ± SEM over 33 games each). **Right (E2):** end accuracy vs. round budget—the 27B and 397B are flat (0.32–0.36); the 4B peaks at cap 15 then degrades to 0.10.

bigram-recall fallback was added. (3) **Truncated thinking masquerades as behavior.** With a 512-token budget, a hybrid reasoning model's visible "questions" were truncated chain-of-thought; disabling template thinking recovered an 8-round solve. Decoding configuration is part of the environment. (4) **Linear-solvable puzzles are poor probes**, motivating structure-graded puzzle construction.

7 Limitations and Ethics

Keyword extraction and encoders are themselves models (dual-encoder reporting planned). SWOW norms are population averages, not a normative standard: human-path distance is a descriptive coordinate, and H2 is a tested correlation, not an assumption that human-unlike is wrong. The proxy manifold used in this draft is explicitly provisional. Judge biases are mitigated per [7, 8]. Puzzles involve dark themes (content-flagged); contamination is handled by memorization probes and fresh/novel sourcing.

8 Conclusion

Turtle soup turns hypothesis-space search into observable text; trajectory geometry over a human-association anchor turns that text into interpretable behavioral signatures. If H1–H3 hold, "why did the agent fail" becomes a measurement—and gives RL training of Questioners a behaviorally grounded reward-shaping target.

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